

TRIZ Principles and 3D Printing

Lecture 10

Outline

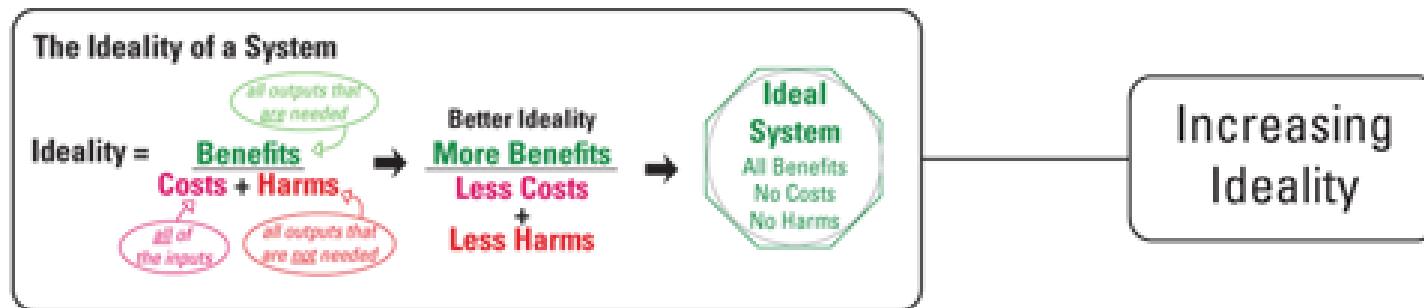
- Trends of Technical Evolution
- 76 standard solutions for dealing with harms and insufficiencies
- Effects database for useful functions
- Introduction to 3D Printing
- Classroom Activity – Fusion 360
 - Animation and Technical Drawing

Trends of Technical Evolution

- Increasing Ideality
- Increasing Segmentation
- Increasing Dynamism and Controllability
- Increasing System Coordination
- Increasing System Automation
- Simplicity – Complexity – Simplicity
- Better Matching of Needs

Increasing Ideality

Systems improve over time by providing more benefits with fewer inputs/costs and resulting in fewer harms.

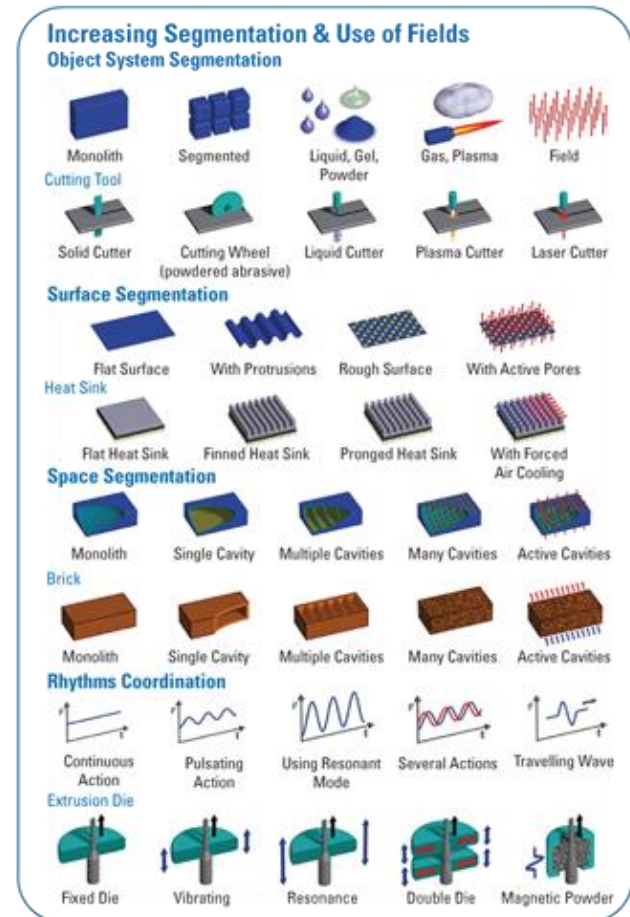


Increasing Segmentation

- Systems become segmented by being broken into ever-smaller pieces, until they become a field effect.

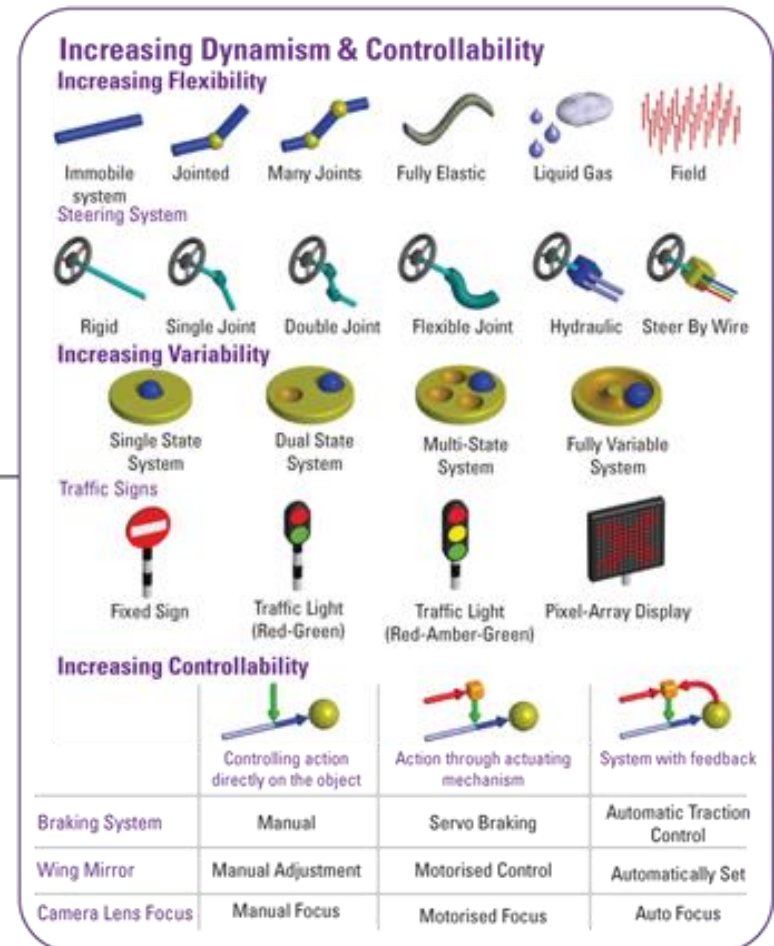


Methods to clean clothes



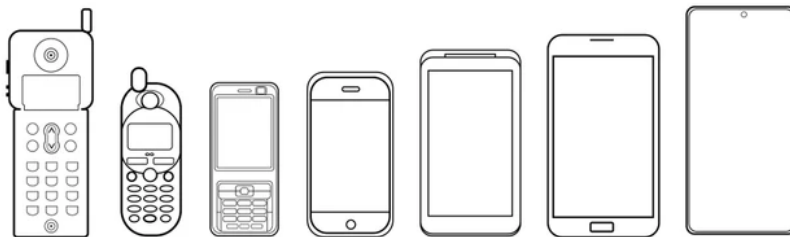
Increasing Dynamism and Controllability

- System components work together more dynamically by becoming more flexible and variable
- Systems increase in controllability to manipulate object via indirect means

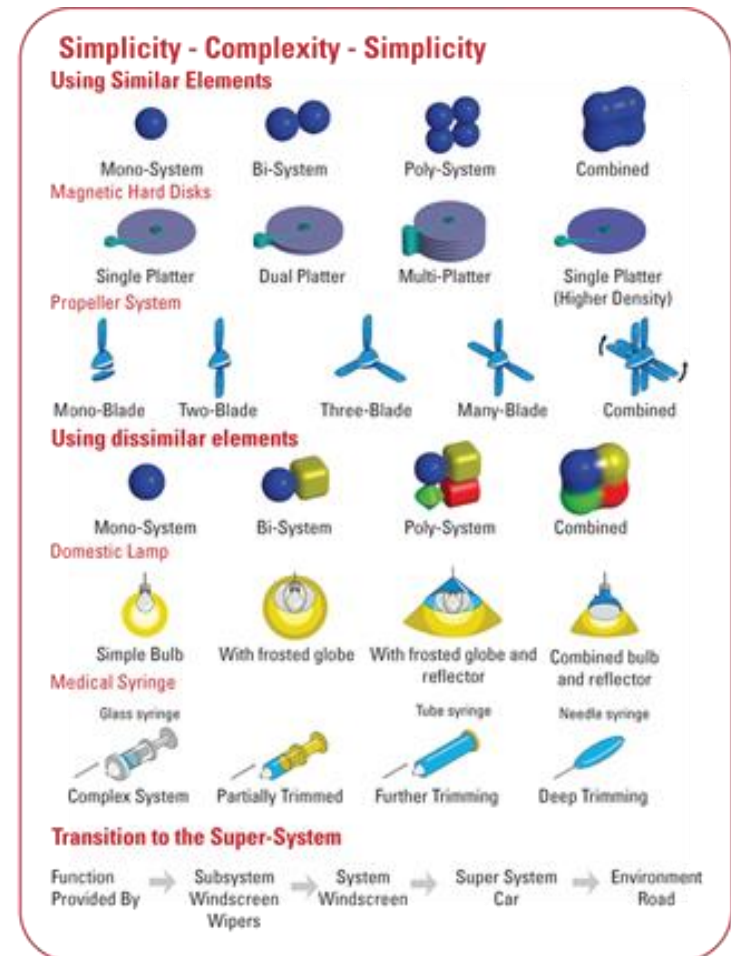


Simplicity – Complexity - Simplicity

- Systems follow a cyclical trend whereby they start simple, become more complex as new elements are added and then become simple again as a result of integrating those new components

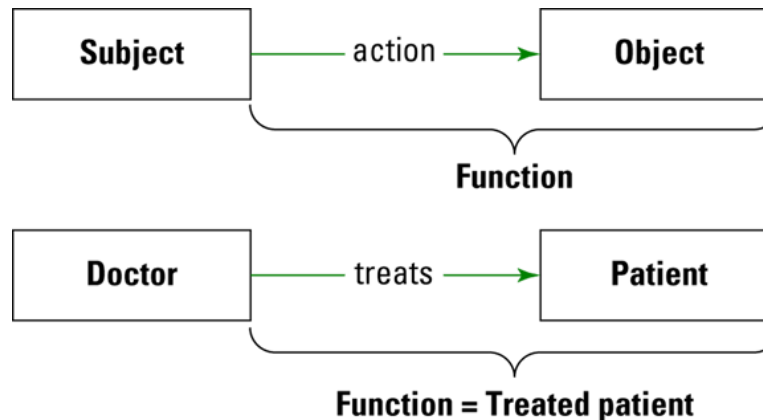


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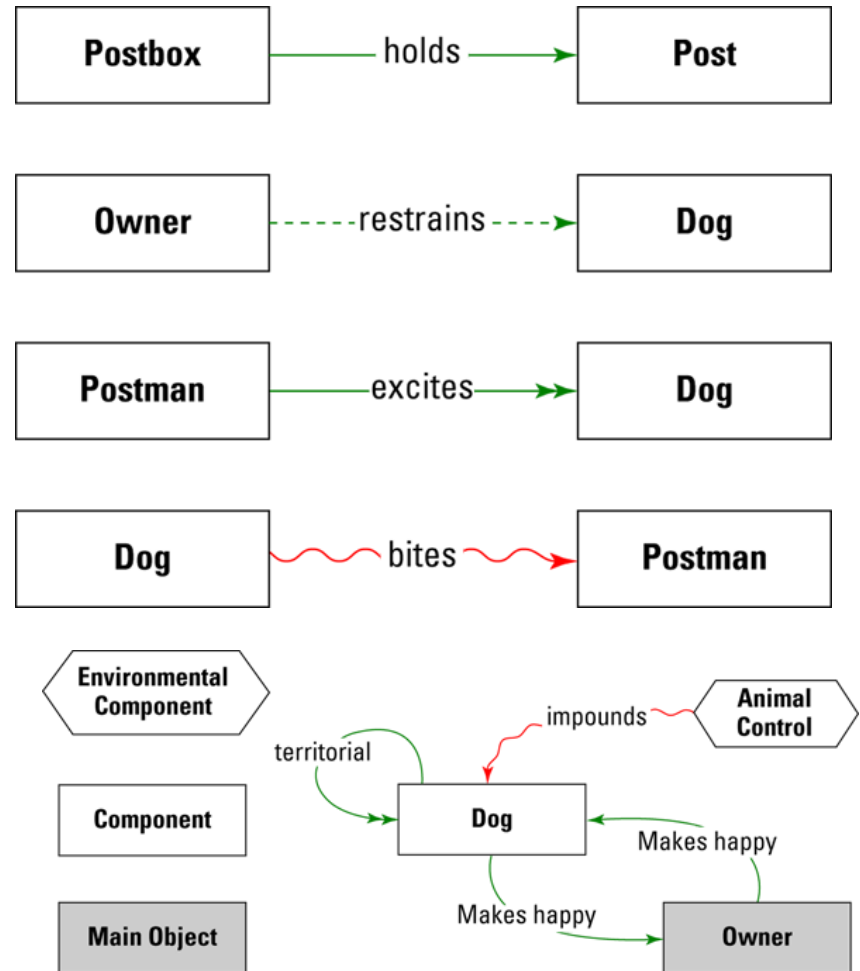
Functional Analysis

- **Subject:** A component which is delivering an action (e.g. microwave oven)
- **Action:** A change being made or a 'doing word' (e.g. heats)
- **Object:** A component which is changed by the action (e.g. food)

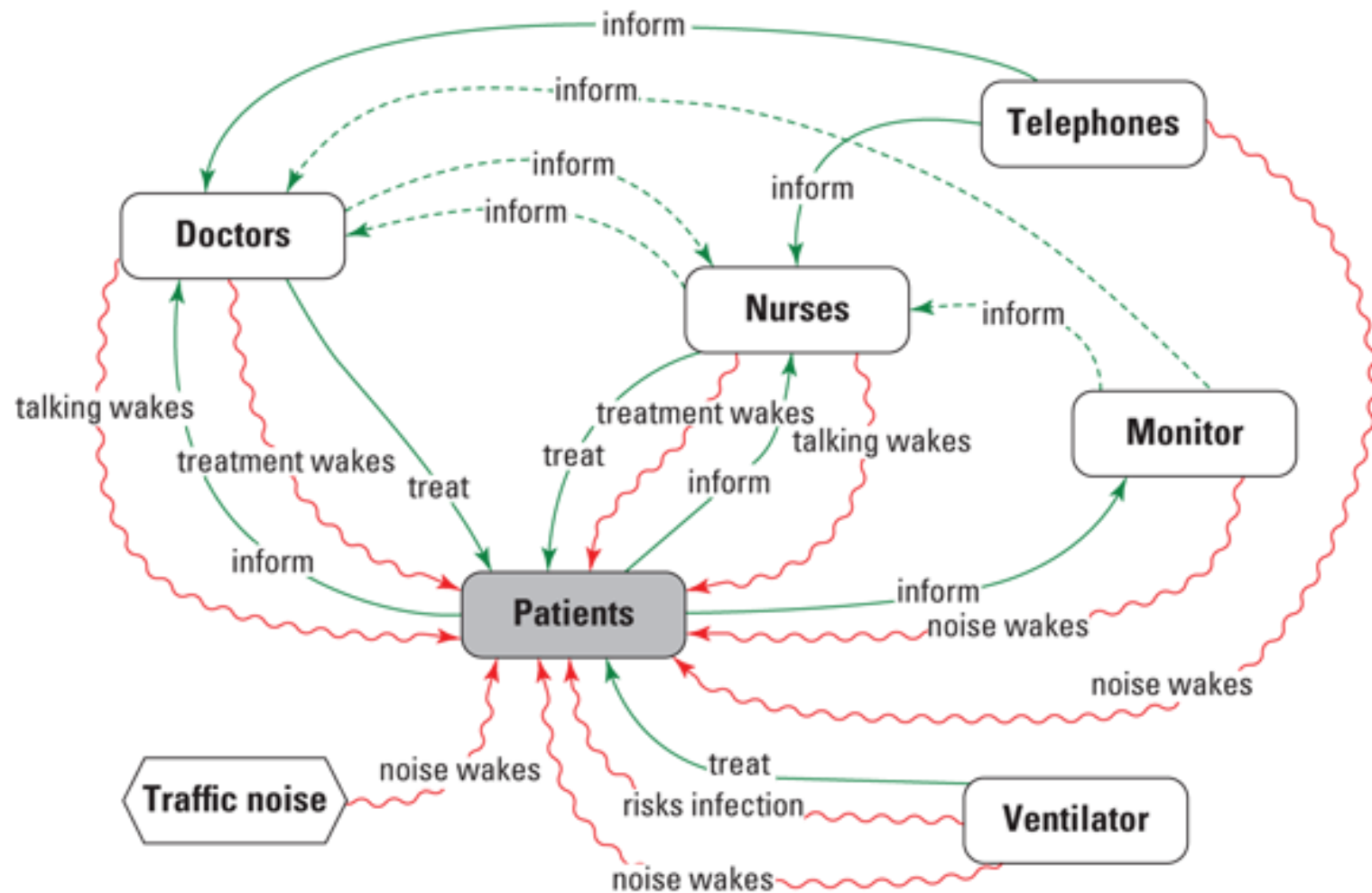


4 kinds of actions and enviroment

- **Useful actions** which deliver something you want
- **Insufficient useful actions** which deliver something you want but not enough of it
- **Excessive useful actions** which delivery too much of something you want
- **Harmful actions** which deliver something you don't want



Function Map of Hospital



Dealing with Harmful Actions

- Trim the harm
 - When useful function can be performed by the object, other component or resource, or can be disposed after use.
- Block the harm
 - Stop the harm having an counterbalancing effect
- Turn the harm into a good
 - Find a way to get a benefit from the harm
- Correct the harm
 - Accept the harm will happen, then correct its effects afterwards

Dealing with Harm Examples

- Trim the Harm
 - H1.5 Could we trim the subject after it has performed its useful action? If yes, trim the subject in time.
 - For example, remove cartridge after bullet is shot.
- Block the Harm
 - H2.2 Change the object so it is not sensitive to the harmful actions.
 - For example, add vents in an umbrella so wind doesn't make it turn inside out.
- Turn the Harm into Good
 - H3.1. Use the harm to deal with the harm.
 - For example, vaccination (small component of a virus stimulates immune response); deliberately burning a strip of land ahead of a wildfire.
- Correct the Harm
 - H4.3 Predict and eliminate harms before they happen, use something that disappears or becomes part of the system/environment.
 - For example, prevent potential damage from removing stitches by using dissolvable stitches in wounds.

76 Standard Solutions for Dealing with Harm or Insufficiencies

Dealing with Harm

H1 = Trim out the harm (6 solutions, so H1.1, H1.2 and so on until H1.6)

H2 = Block the harm (11 solutions)

H3 = Turn harm into good (4 solutions)

H4 = Correct the harm afterwards (3 solutions)

Overcoming Insufficiency

i1 = Add something to the subject or object (7 solutions)

i2 = Evolve the subject and object (10 solutions)

i.a. = Improve the action (18 solutions)

Difficulties with Measuring or Detecting

M1 = Indirect methods (3 solutions)

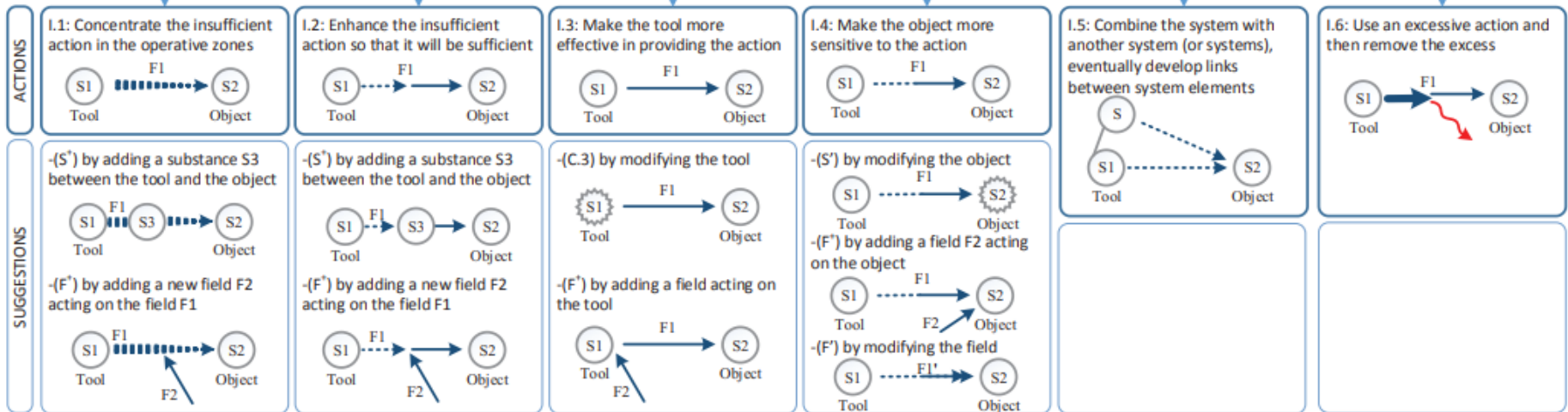
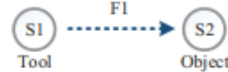
M2 = Add something (4 solutions)

M3 = Enhance measurement with fields (3 solutions)

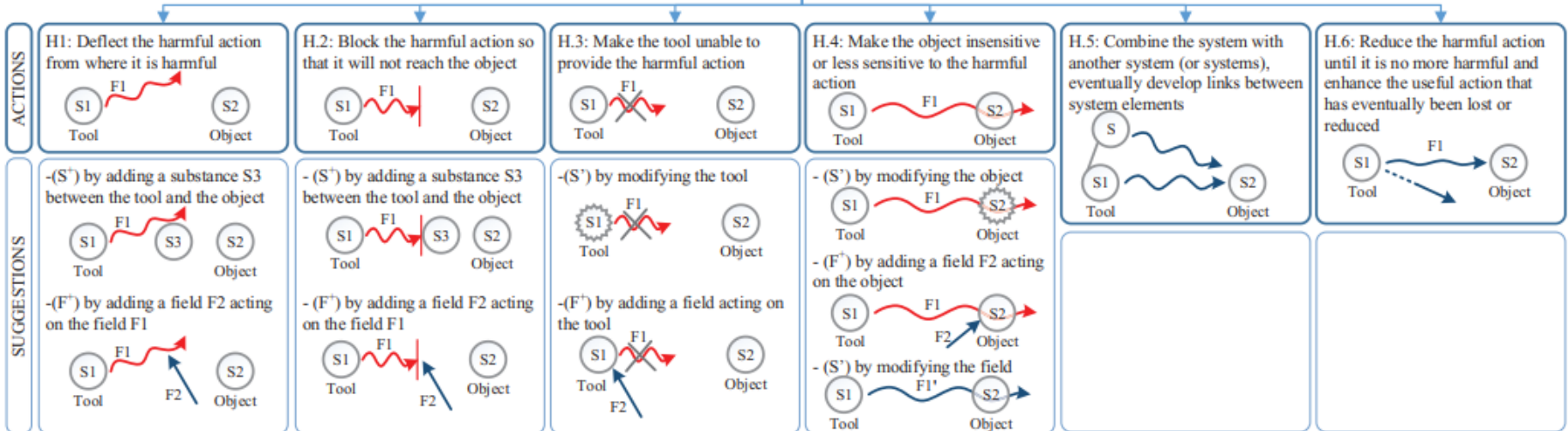
M4 = Use additives with fields (5 solutions)

M5 = Evolve the measurement system (2 solutions)

I: INSUFFICIENT AND MISSING ACTIONS

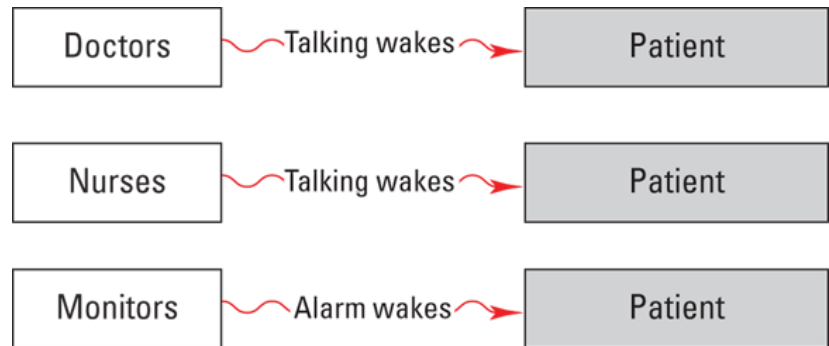


H: HARMFUL AND EXCESSIVE ACTIONS



Hospital Example

- H2.1 Counteract the noise with an opposing field that neutralises the noise
 - White noise, or ocean sounds improves sleep by about 40%.
- H2.3 Change the zone and duration of the noise to decrease its effects
 - Implement quiet times during the day and the night when no visitors, radios or TVs allowed.
- H2.4 Insulate from the noise by introducing a new component or substance
 - Ear muffs or ear plugs, individual private wards
- H2.9 Use sub-system components to stop the noise
 - Designing devices with other types of alarms or ventilation, use pagers for doctors communication
- H2.10 Use super-system components to stop the noise
 - Soundproofing walls, lowering ceiling to absorb ambient noise.



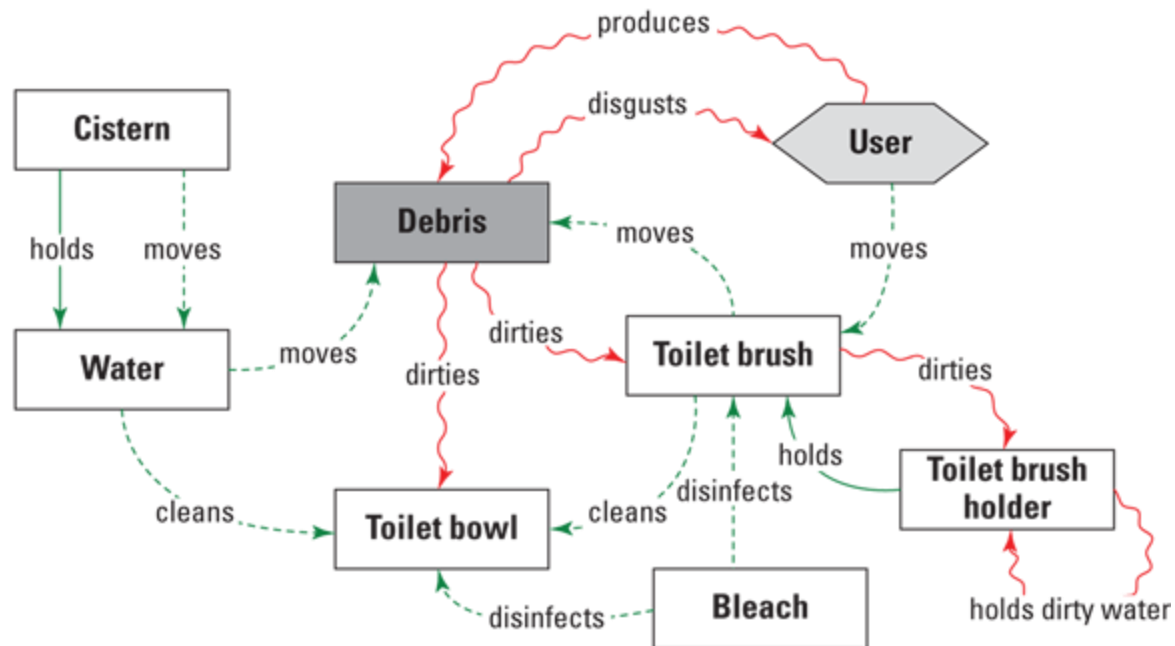
Improving Insufficiency Examples

- Add something to the subject or object
 - i1.2 Add something between the subject and object which enhances the function.
 - For example, spectacles to enhance eyes; wrapping baking potato in foil to speed up cooking;
- Evolve the subject and object
 - i2.8 Transition function delivery to the micro-level.
 - For example, self-cleaning glass; microcapsules in cosmetics; microwave ovens
- Improve the action
 - i.a.2 Add another action from existing components.
 - For example, sonic toothbrushes use sound waves as well as mechanical vibration to clean teeth.

Detection and Measurement Examples

- Indirect methods
 - M1.1 Change the problem so there is no need for measurement or detection.
 - For example, cooking in boiling water doesn't require measuring temperature: boiling water is always at 100°C.
- Add something
 - M2.1 Add something and then measure changes to one of its features.
 - For example, there are egg timers that you place in the same saucepan of water as your egg, which change colour to tell you when the egg is cooked; car tyre tread-wear indicators.
- Evolve the measurement system
 - M5.1 Go from one measurement to two or more to improve the quality of relying on a single measurement.
 - For example, measure temperature in multiple places or times.

Toilet Brush Example



Resolving Insufficiencies

- I1.1 Add something to or inside the toilet brush or debris
 - This could suggest making the toilet brush handle hollow, and adding bleach or some other detergent inside the brush that's automatically dispensed in use
- I1.3 Use the external environment to enhance/provide the function of move debris
 - Encourage the debris to move by vibrating the toilet bowl
 - Adding a cleaning setting to toilet which dispenses a small amount of water from the cistern to aid the cleaning process.
 - Non-stick toilet bowl
- I1.4 Change or add something to environment of the toilet brush and debris
 - Add some chemical to the water in cistern that aides in loosening the debris
- I2.1 Segment the toilet brush or debris: increase the degree of fragmentation or divide into smaller units
 - Small fragmented bristles on top to improve the cleaning action

Using the TRIZ Effects Database

- **TRIZ Effects Database** is a catalogue of all the known scientific and engineering effects discovered so far, arranged into simple questions and answers.
- Steps of Using Effects Database:
 1. Describe your problem in clear language such as “How?” question. For example, “How do I purify water?” or “How do I ensure water is clean?”
 2. Define an X-Factor question to describe what you are hoping to do.
 - Function examples: clean a liquid, constrain a gas, detect a solid, etc
 - Parameter examples: measure purity, increase friction, change density, etc.
 3. Go to Effects Database (<http://wbam2244.dns-systems.net/EDB/>) and find required effect
 4. Translate conceptual solution into your product domain

Introduction to 3D Printing

- **3D printing**, or **additive manufacturing**, is the construction of a three-dimensional object from a CAD model or a digital 3D model.
- The term "3D printing" can refer to a variety of processes in which material is *deposited*, *joined* or *solidified* under computer control to create a 3D object, with material being added together typically layer by layer.
- Use of 3D Printing:
 - Product Design and Prototyping
 - Architecture
 - Medical and Prosthetics
 - STEM education

Advantages of 3D printing

- **Less waste**

- In traditional '*subtractive*' manufacturing, a block of material is cut down to create an object, which can result in up to 90% in waste material.
- Because 3D printing uses an additive process, little or no material is wasted.

- **Customization**

- Products such as pens, phone cases etc are created on assembly lines because it is cost efficient to produce them in this way.
- With 3D printing, each design can be different and there will be no additional tooling or assembly costs involved.

Advantages of 3D printing

- **Complex Geometries**

- Subtractive manufacturing techniques have limitations in the geometry of products.
- Machines simply cannot cut or mill certain patterns but with 3D printing, these limitations don't exist.

- **Efficient Prototyping**

- Imagine being able to design a concept and hold a prototype of this within a matter of hours.
- 3D printing enables this at just a fraction of the cost of creating prototypes through traditional methods.

Disadvantages of 3D printing

- **Speed of Large Production Runs**

- 3D printing is a great tool to obtain prototypes and low quantities of products quickly because there is no setup required.
- For large production runs, 3D printing is not a viable option because of the speed of the actual printing process.

- **Cost of Large Production Runs**

- In traditional manufacturing, the cost of production decreases with the amount of products being made.
- With 3D printing, this is not the case because cost is based on material usage.

Disadvantages of 3D printing

- **Limited Materials**

- More and more new materials are becoming available for 3D printers but there is still a very long way to go before there are anywhere near the amount of options available through traditional methods.

- **Accuracy**

- In some industries such as engineering and automotive, small components require extreme precision.
- Although 3D printers can be precise right down to around 10 microns, subtractive methods can still achieve better precision.

3D printing steps



1. 3D model file

It all begins with a digital 3D model of a design. There are hundreds of software programs that enable you to design in 3D. Some free ones that are great for educators are SketchUp, TinkerCad and Fusion 360.



2. Slicing

The 3D model file (typically an STL file) is then split up into very fine layers in a software program called a “slicer”. The output from the slicer is a code, which tells the 3D printer how to move and where to lay down material.



3. 3D Printing

Once the code is loaded to the 3D printer, it will begin the production process. One layer is created at a time until the model is complete. This can take minutes or hours depending on the size of the model.



4. Post-processing

Depending on the type of 3D printer and the model file you print, there may be a requirement for some post-processing. Examples include removing support material, cleaning and sanding.

3D printing types



1. FFF (Fused Filament Fabrication)

This is the most popular type of 3D printer for educators and they work by melting plastic filament and depositing it onto a print bed, where it solidifies. Additional layers are printed on top of each other until the model is complete. Types of plastic include PLA, ABS and composites that combine plastic with wood, copper, bronze and several others.



2. SLA/DLP (Stereolithography/Digital Light Processing)

SLA/DLP machines utilise light sources to cure a liquid photopolymer. A print bed is submerged in resin and the light is used to solidify certain areas. Once a layer has solidified the print bed will move, allowing the light to cure the next layer.



3. SLS (Selective Laser Sintering)

Selective Laser Sintering works in a similar manner to SLA/DLP technologies but the light source (laser) is used to fuse powdered material together. The most common material for SLS printers is nylon.



4. Material Jetting

Material jetting uses a similar technology to a standard inkjet printer. The difference being that the material used is a liquid photopolymer that is cured by a UV light source once it hits the print bed.

Fused filament fabrication (FFF) – Material Extrusion

- How it works?

1. Plastic filament is extruded with a motor through a heated nozzle
2. Nozzle deposits the material on a build platform along predetermined path
3. Filament cools and solidifies making a 3D object layer by layer

- Filament types:

- PLA, ABS, PET, PETG, TPU, Nylon, ASA, PC, HIPS, Carbon Fiber ([more here](#))

- Dimension Accuracy:

- +/- 0.5%

- Common Applications:

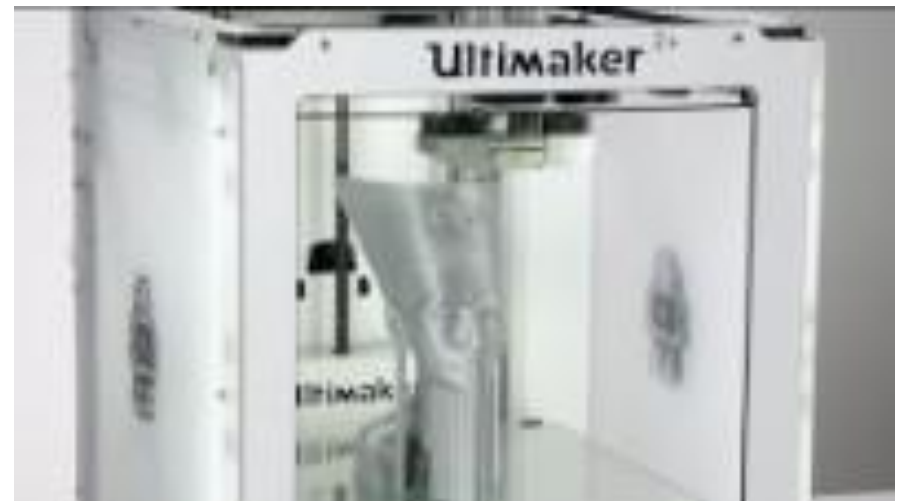
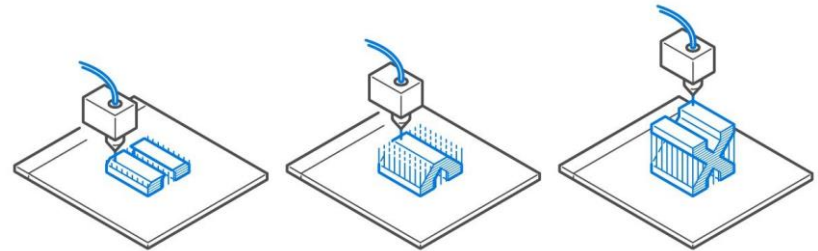
- Electrical housings, form and fit testing, figs and fixtures

- Pros:

- Low cost of printer (around \$250 to \$25k)
- Low cost of material
- Wide variety of materials

- Cons:

- Tiny dimensions 10x10x10 to 60x60x60 cms
- Non-durable
- Extra support for hanging parts



Filament Types – Polylactic Acid (PLA)

Strength	High
Flexibility	Low
Durability	Medium
Print temperature	180 – 230 °C
Print bed temperature	20 – 60 °C
Odour	Minimal
Shrinkage/warping	Minimal
Soluble	No
Food safety	Depends
Used for:	Low-wear toys, prototype parts and containers
Don't use for:	Items that bend, twists or wear often. Items that withstand high temperature or high mechanical force.



Filament Types – Acrylonitrile butadiene styrene (ABS)

Strength	High
Flexibility	Medium
Durability	High
Print temperature	210 – 250 °C
Print bed temperature	80 – 110 °C
Odour	Intense
Shrinkage/warping	Considerable
Soluble	In esters, ketones and acetones
Food safety	Not safe
Used for:	High-wear toys, tool handles. Heat and force resistant items.
Don't use for:	Prototyping parts or low-wear toys



Filament Types – Polyethylene terephthalate (PETG)

Strength	High
Flexibility	Medium
Durability	High
Print temperature	220 – 250 °C
Print bed temperature	50 – 75 °C
Odour	Minimal
Shrinkage/warping	Minimal
Soluble	No
Food safety	Depends
Used for:	Flexible and durable items
Don't use for:	Scratch resistant items, items with hanging parts (which requires support)



Filament Types – thermoplastic elastomers (TPE, TPU and TPC)

Strength	Medium
Flexibility	Very high
Durability	Very high
Print temperature	210 – 230 °C
Print bed temperature	30 – 60 °C
Odour	Minimal
Shrinkage/warping	Minimal
Soluble	No
Food safety	Depends
Used for:	Anything that bends, stretches or compresses.
Don't use for:	Requires tight filament path and slow printing



Other Filaments

Nylon

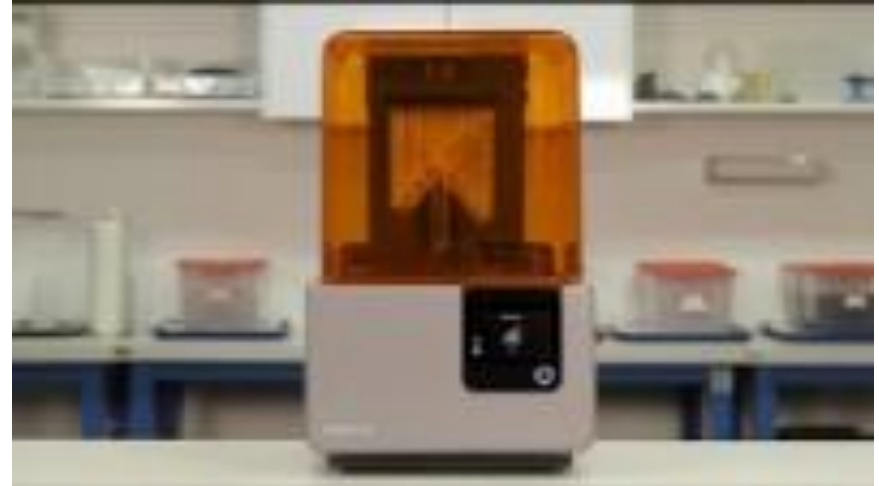
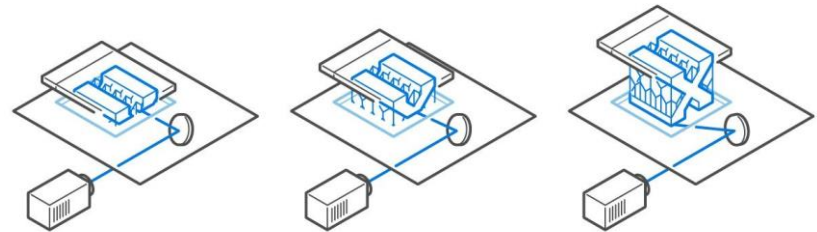
- Used for mechanical parts such as hinges, buckles and gears
- Pros:
 - High strength
 - High flexibility
 - High durability
- Cons:
 - Expensive
 - Susceptible to moisture
 - Requires high nozzle
 - Requires bed temperature

Polycarbonate (PC)

- Used for tough and high-temperature environments such as electrical, mechanical and automotive parts.
- Pros:
 - Extremely strong
 - Resistant to heat
 - Resistant to physical impact
- Cons:
 - Susceptible to moisture
 - Requires very high print temperature

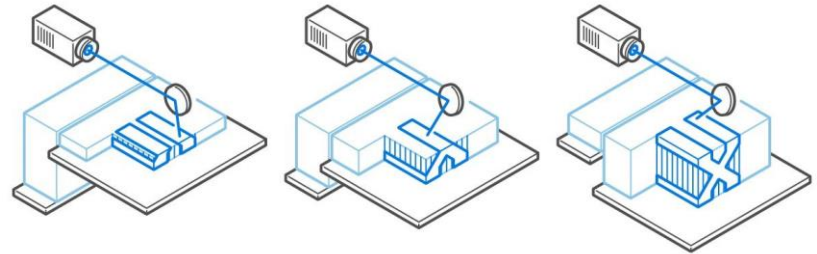
Stereolithography (SLA) – Vat polymerization

- How it works?
 1. light source selectively cures a photopolymer resin in a vat
 2. this process is repeated layer by layer until the 3D part is formed.
- Materials:
 - Photopolymer resin
- Dimension Accuracy:
 - +/- 0.5% (lower limit 0.15 mm)
- Common Applications:
 - Mold-like polymer prototypes, jewelry casting, dental applications
- Pros:
 - Smooth surface finish
 - Fine feature details
- Cons:
 - Tiny dimensions 10x10x10 to 60x60x60 cms
 - Expensive printer and materials



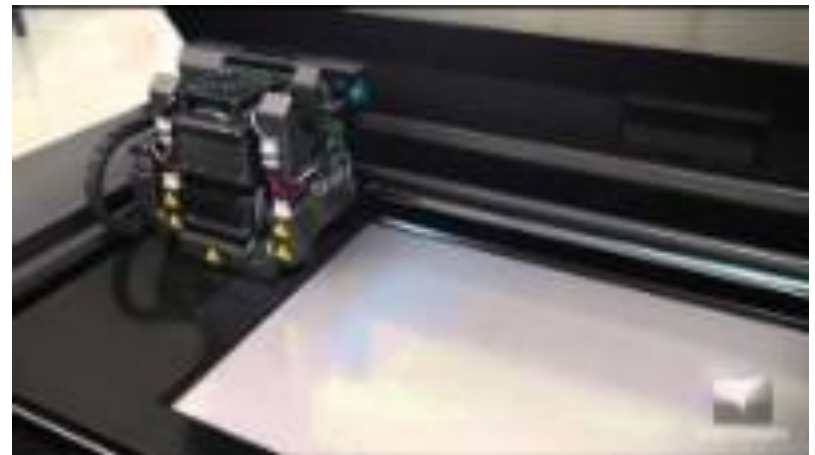
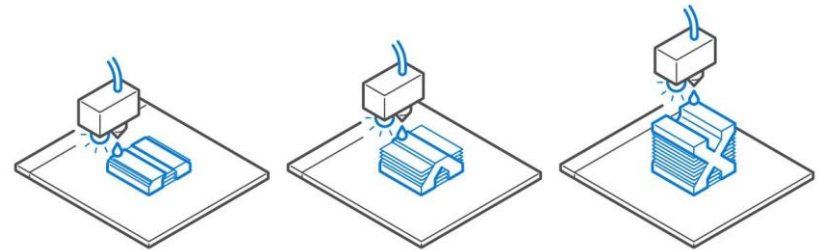
Selective Laser Sintering (SLS) – Powder bed fusion

- How it works?
 1. thermal energy source selectively induces fusion between powder particles (plastic, metal, or ceramic) inside a build area
 2. this process is repeated layer by layer until the 3D part is formed.
- Materials:
 - Thermoplastic powders (Nylon 6, Nylon 11, Nylon 12, etc.)
 - Metal powders (steel, titanium, aluminum, cobalt)
 - Ceramic powders
- Dimension Accuracy:
 - +/- 0.3% (lower limit 0.3 mm)
- Common Applications:
 - Function parts of mechanical hardware
- Pros:
 - Excellent mechanical properties
 - Complex geometries
- Cons:
 - Tiny dimensions 10x10x10 to 60x60x60 cms
 - Expensive printer and materials



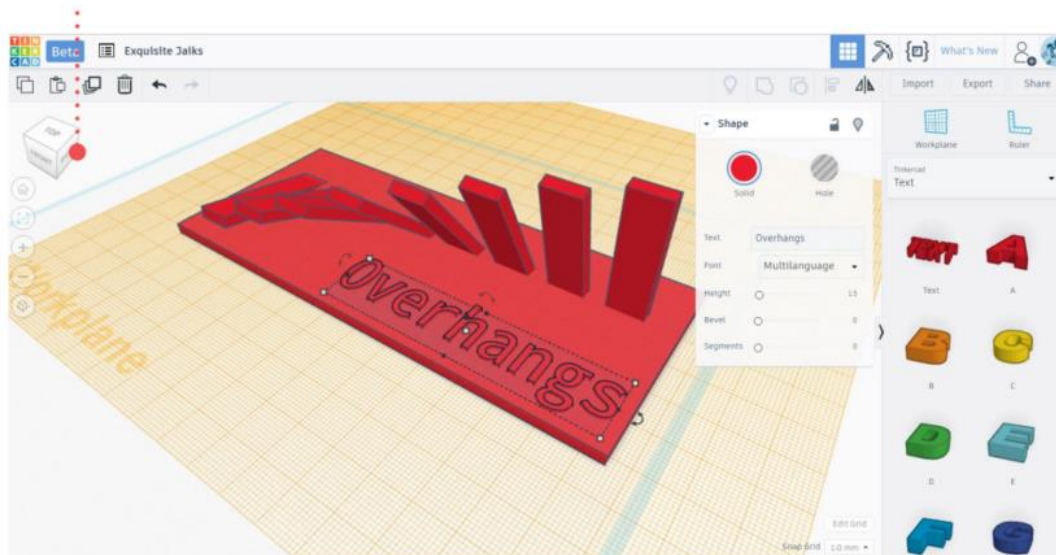
Material Jetting

- How it works?
 1. photopolymers or wax droplets are selectively deposited on a build plate.
 2. when exposed to light droplets cure and objects are built up, one layer at a time.
- Materials:
 - Photopolymer resin (Standard, Castable, Transparent, High Temperature)
- Dimension Accuracy:
 - ± 0.1 mm
- Common Applications:
 - Full-color product prototypes; Injection mold-like prototypes; Low run injection molds; Medical models
- Pros:
 - Best surface finish;
 - Full color and multi-material available
- Cons:
 - Brittle, not suitable for mechanical parts;
 - Higher cost than SLA/DLP for visual purposes



CAD Software

- CAD stands for Computer-Aided Design Software and refers to computer programs that are intended for design purposes.
- They can either be 2D, 3D or both and most 3D printable files are created with CAD software.

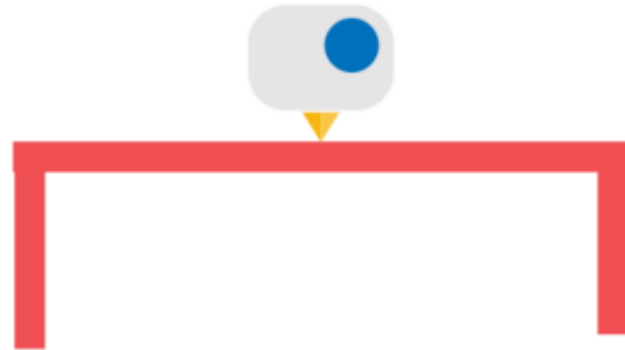


CAD Software Options

- Tinkercad – www.tinkercad.com
 - Simple, online 3D design and 3D printing app that can be used through a web browser
 - Great for beginners with an onboarding tutorial.
- SketchUp – www.sketchup.com
 - Simple, offline/online software.
 - 2D sketches can be extruded into 3D shapes.
 - Great for beginners and intermediate.
- Fusion 360 - www.autodesk.com/products/fusion-360
 - Fusion 360 is a cloud-based 3D CAD, CAM, and CAE platform for product development.
 - It combines industrial and mechanical design, simulation, collaboration and machining in a single package.
 - Advanced design software for professionals

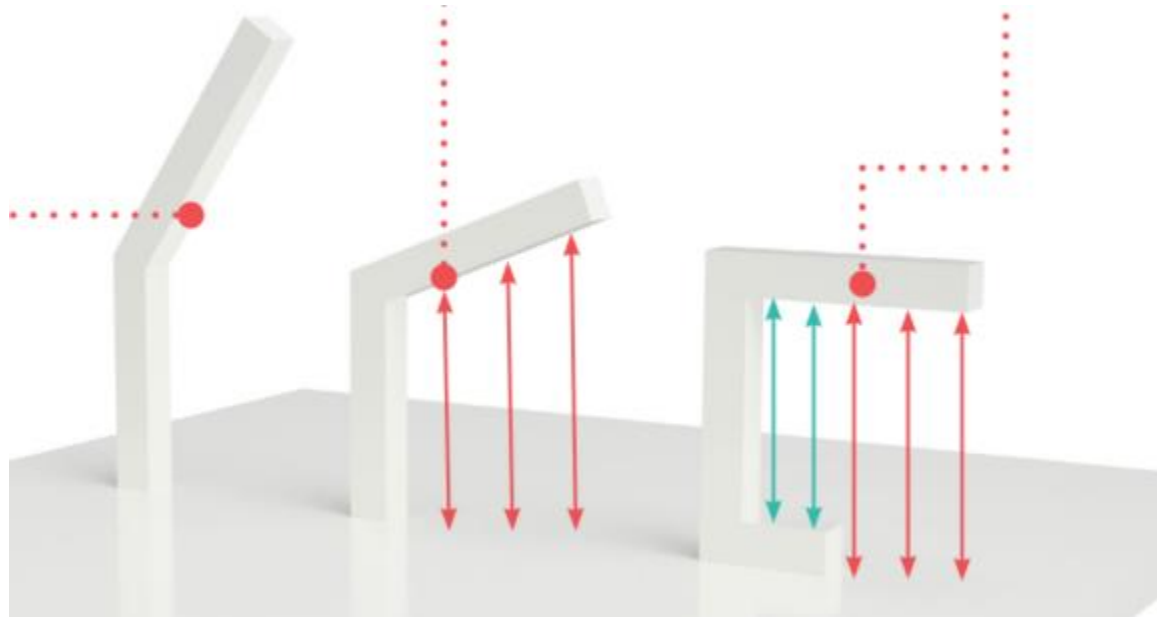
Design Considerations for 3D Printing

Overhangs

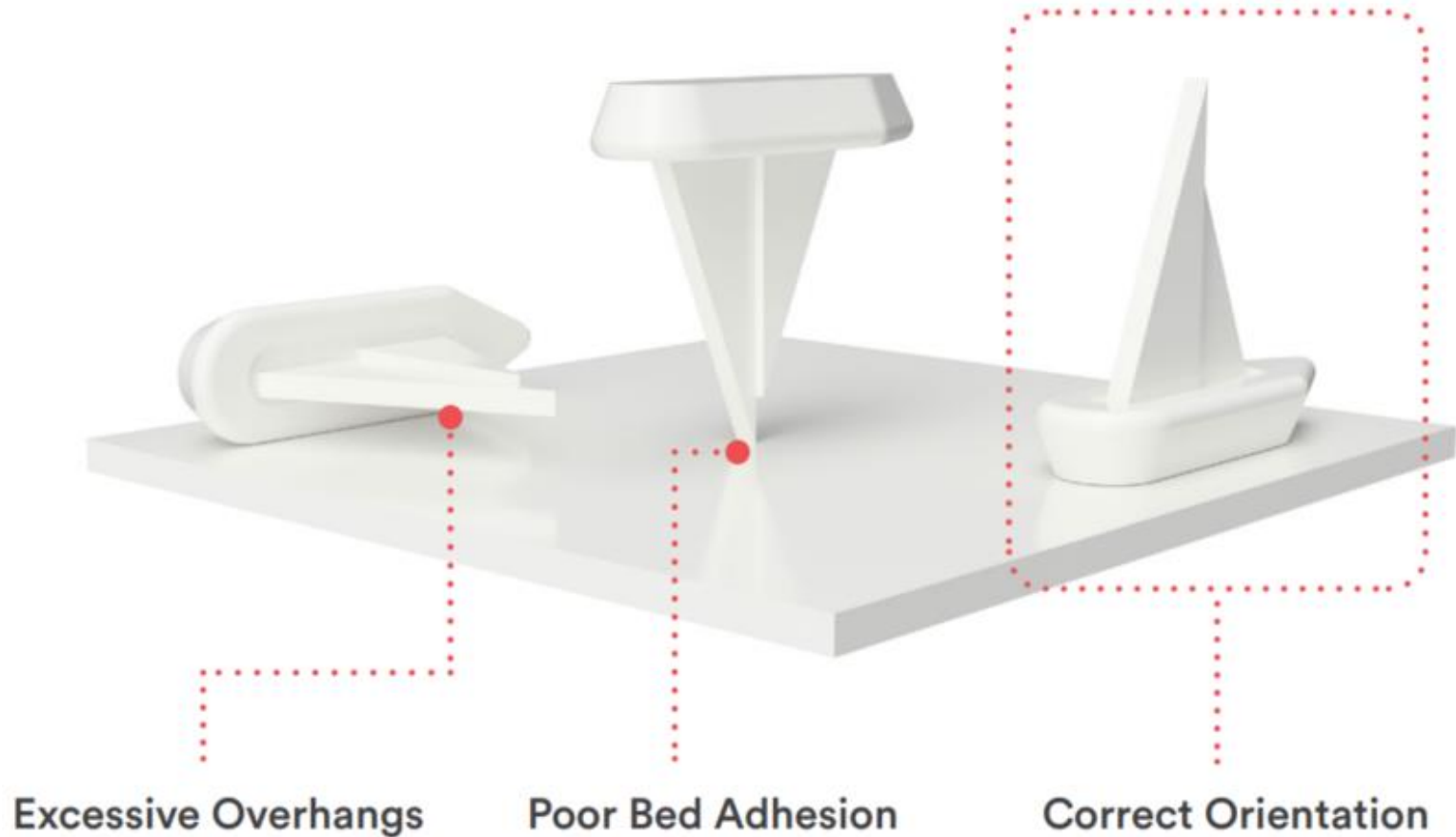


Supports are generated by most slicers.

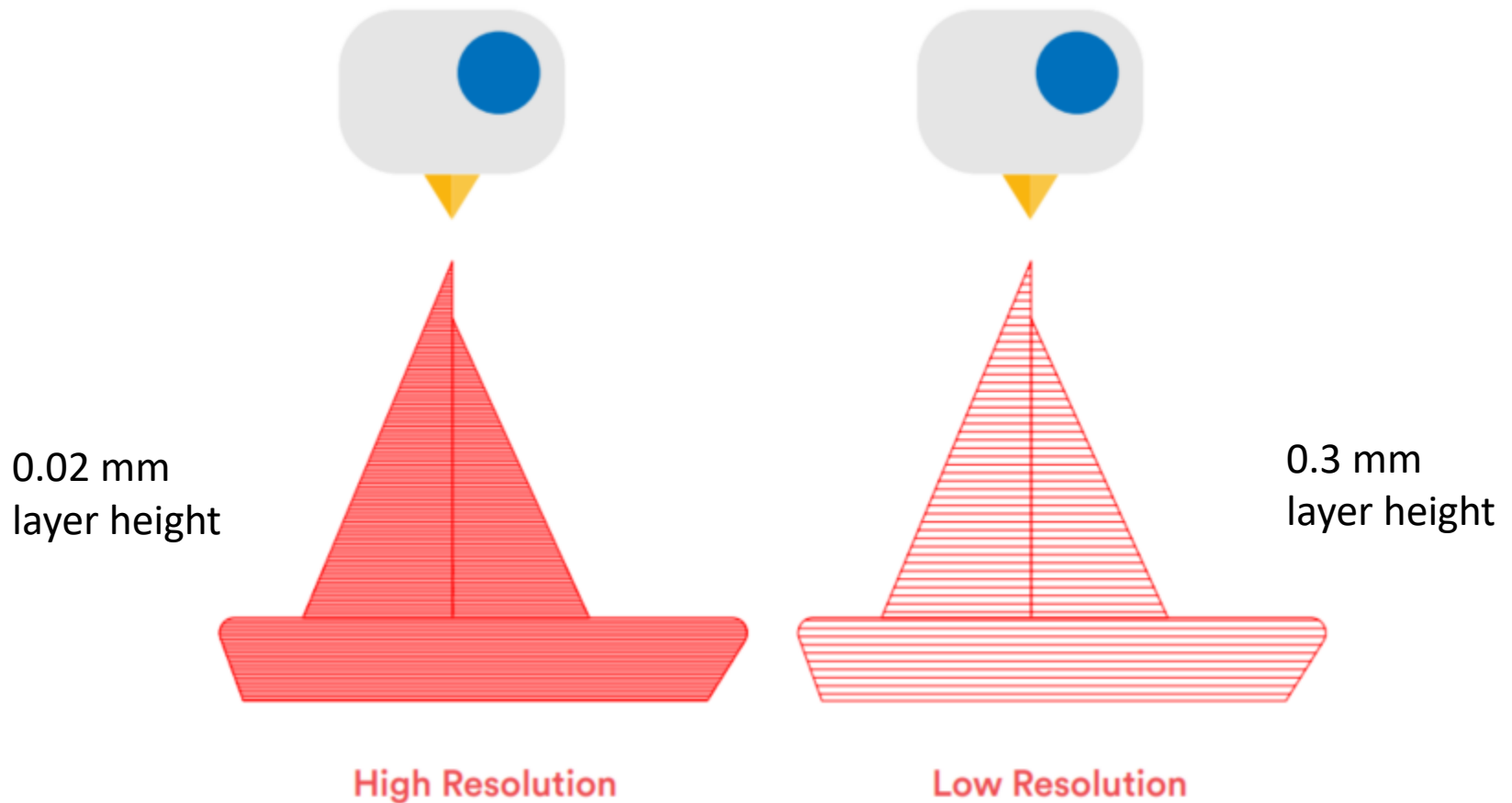
Design Considerations for 3D Printing



Design Considerations for 3D Printing



Design Considerations for 3D Printing

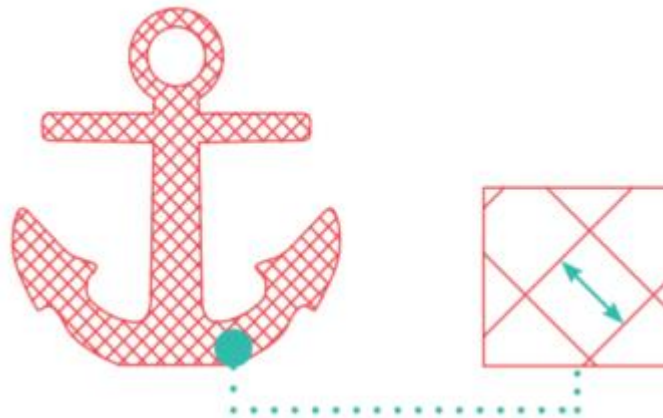


Design Considerations for 3D Printing

Hollow



Regular – 4-6 mm apart,
or 20% density



Solid



Useful Resources

- Online video tutorials
 - <https://help.autodesk.com/view/fusion360/ENU/courses>
 - <https://help.autodesk.com/view/fusion360/ENU/courses/AP-MFG-ADD-F360-INTRO>
 - <https://help.autodesk.com/view/fusion360/ENU/courses/AP-MFG-ADD-FFF>
- 3D models library:
 - www.thingiverse.com
 - www.youmagine.com
 - www.myminifactory.com
- 3D printing marketplaces in Uzbekistan
 - <https://3dbozor.uz>
 - https://www.instagram.com/3d4p_tashkent
 - <https://unit3d.uz/services/printing>
 - <https://t.me/at3dprint>
 - <https://t.me/iqshopuz>
- Injection molding companies in Uzbekistan (for mass manufacturing)
 - <https://lochinmould.com>
 - <http://sps.uz>
 - <https://www.meridian.uz>
 - <https://www.prom.uz/section/izgotovlenie-press-form-dlya-litya-plastmass>

More on TRIZ

- Samples that the students can try
 - <https://triz.org/principles/>
- More TRIZ resources:
 - <https://www.triz.co.uk/>